Cycle Test 1: worked solutions

20 Aug 2025 · 20 marks · 1 hour · CEPC22

Q1

2 marks

> Define: (a) specific weight (b) viscosity (c) surface tension (d) capillarity.

- **(a) Specific weight (w)**: weight per unit volume of a fluid, $w = \rho g$ (N/m^3). Water: 9810 N/m^3.
- **(b) Viscosity**: the property of a fluid by which it **resists the relative motion (shear) between adjacent layers**. Dynamic viscosity μ is defined by Newton's law, $\tau = \mu \, du/dy$ (unit N·s/m^2 = Pa·s; 1 poise = 0.1 N·s/m^2).
- **(c) Surface tension (σ)**: the tensile force per unit length acting along the free surface of a liquid, caused by cohesion between molecules at the surface (N/m). Water–air at 20 °C: 0.0725 N/m.
- **(d) Capillarity**: the **rise or fall of a liquid in a small-diameter tube** dipped in it, due to surface tension and adhesion/cohesion. Rise $h = 4\sigma \cos\theta / (\rho g d)$. Water rises (adhesion > cohesion); mercury falls.

Q2

2 marks

> List the types of fluid flow and explain any three in brief.

Types: steady and unsteady; uniform and non-uniform; laminar and turbulent; compressible and incompressible; rotational and irrotational; one-, two- and three-dimensional.

- **Steady flow**: fluid properties (velocity, pressure, density) at a point **do not change with time**: $\partial V/\partial t = 0$. *Unsteady*: they change with time.
- **Uniform flow**: velocity **does not change with distance** along the flow at a given instant: $\partial V/\partial s = 0$ (e.g. flow in a long pipe of constant diameter). *Non-*

uniform: velocity changes along the path (e.g. tapering pipe).

- **Laminar flow**: particles move in smooth, parallel layers (laminae) without mixing; Re < 2000 in pipes. *Turbulent*: particles move irregularly with mixing; Re > 4000.
- **Rotational / irrotational**: fluid particles rotate about their own axis / do not rotate ($\omega = 0$).

Q3

2 marks

> A flat plate of area 1.5×10^6 mm^2 is pulled with a speed of 0.4 m/s relative to another plate located 0.15 mm from it. Find the force (N) and power (W) required to maintain this speed if the fluid separating them has viscosity 0.1 N·s/m^2.

$A = 1.5 \times 10^6$ mm^2 = **1.5 m^2**; $du = 0.4$ m/s; $dy = 0.15 \times 10^{-3}$ m.

$\tau = \mu\, du/dy = 0.1 \times 0.4 / (0.15 \times 10^{-3})$ = **266.67 N/m^2**

Force $F = \tau \times A = 266.67 \times 1.5 =$ **400 N**

Power $P = F \times u = 400 \times 0.4 =$ **160 W**

Q4

2.5 marks

> State Pascal's law and prove it.

Pascal's law: the **intensity of pressure at a point in a fluid at rest is the same in all directions**.

Proof: consider a small wedge-shaped element ABC of the fluid at rest, of unit width, with faces AB (vertical, height dy), BC (horizontal, width dx) and AC (inclined, length ds, at angle θ). Let p_x, p_y and p_z be the pressures on the vertical, horizontal and inclined faces. A fluid at rest has **no shear stress**, so only pressure forces (normal to faces) and weight act.

- Horizontal equilibrium: $p_x \times dy \times 1 = p_z \times ds \times 1 \times \sin\theta$. Since $ds \sin\theta = dy$, **$p_x = p_z$**.

- Vertical equilibrium: $p_y \times dx \times 1 = p_z \times ds \times 1 \times \cos\theta$ + weight ($\tfrac{1}{2}\, dx\, dy \times 1 \times \rho g$). Since $ds\cos\theta = dx$, and the weight term is of second order ($\to 0$ as the element shrinks to a point), $\mathbf{p_y = p_z}$.

Hence $\mathbf{p_x = p_y = p_z}$. Since θ is arbitrary, pressure at a point is the same in every direction.

Q5

2.5 marks

> A stream function in a two-dimensional flow is $\psi = 2xy$. Show that the flow is irrotational and determine the corresponding velocity potential ϕ.

Using $u = \partial\psi/\partial y$, $v = -\partial\psi/\partial x$:

- $u = \partial(2xy)/\partial y = \mathbf{2x}$
- $v = -\partial(2xy)/\partial x = \mathbf{-2y}$

Irrotational check: $\omega_z = \tfrac{1}{2}(\partial v/\partial x - \partial u/\partial y) = \tfrac{1}{2}(0 - 0) = \mathbf{0} \to$ the flow is **irrotational** (equivalently, $\nabla^2\psi = \partial^2\psi/\partial x^2 + \partial^2\psi/\partial y^2 = 0 + 0 = 0$).

Velocity potential ($u = \partial\phi/\partial x$, $v = \partial\phi/\partial y$):

- $\partial\phi/\partial x = 2x \to \phi = x^2 + f(y)$
- $\partial\phi/\partial y = f'(y) = -2y \to f(y) = -y^2 + C$

$$\boxed{\phi = x^2 - y^2 + C}$$

(With the opposite sign convention, $u = -\partial\psi/\partial y$ and $u = -\partial\phi/\partial x$ as in some Indian textbooks, the same $\phi = x^2 - y^2$ is obtained.)

Q6

3 marks

> Water in a tank is pressurized by air; the pressure is measured by a multi-fluid manometer as shown. The tank is on a mountain at 1400 m where atmospheric pressure is 85.6 kPa. Find the air pressure in the tank if $h_1 = 0.1$ m, $h_2 = 0.2$ m, $h_3 = 0.35$ m. Densities: water 1000, oil 850, mercury 13,600 kg/m^3.

Start at the air–water interface (point 1, pressure P_1) and move through the tube to the open end (point 2, atmospheric). Going **down** adds ρgh; going **up** subtracts it:

$$P_1 + \rho_w g\, h_1 + \rho_{oil} g\, h_2 - \rho_{Hg} g\, h_3 = P_{atm}$$

$P_1 = P_{atm} - \rho_w g h_1 - \rho_{oil} g h_2 + \rho_{Hg} g h_3$

$= 85{,}600 - 1000(9.81)(0.1) - 850(9.81)(0.2) + 13{,}600(9.81)(0.35)$

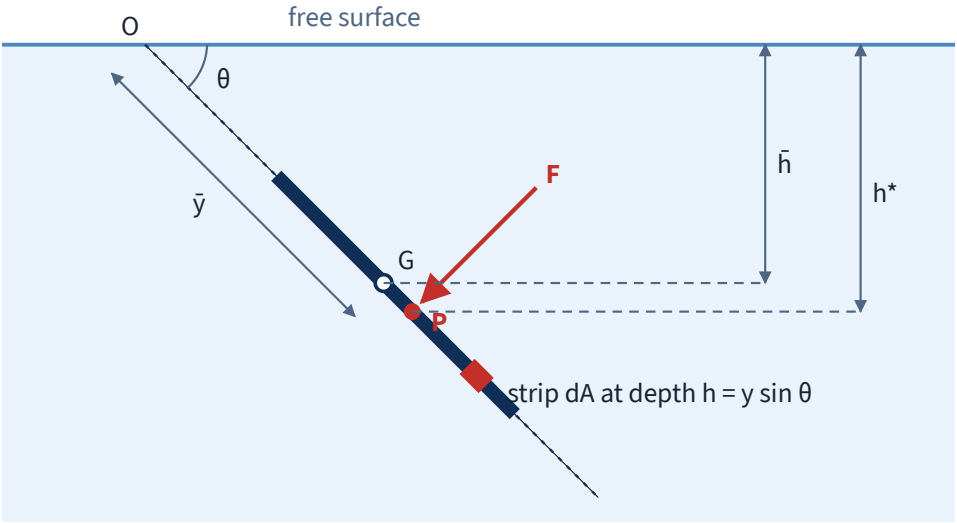
$= 85{,}600 - 981 - 1667.7 + 46{,}695.6$

= **$129{,}647$ Pa ≈ 129.6 kPa (absolute)** = 44.0 kPa gauge

Q7

2 marks

> Derive the expression for centre of pressure on a fully submerged inclined plane surface.



Plane area A inclined at θ to the free surface. G = centroid at depth $\bar{h}$; P = centre of pressure at depth h*. y is measured along the plane from O, where the plane (extended) meets the free surface.

Total pressure. A strip dA at distance y along the plane is at depth h = y sinθ, so the force on it is dF = ρg h dA = ρg y sinθ dA.

$$F = \int \rho g\, y \sin\theta \, dA = \rho g \sin\theta \int y \, dA = \rho g \sin\theta \, (A\,\bar{y}) = \mathbf{\rho g\, A\, \bar{h}}$$

Centre of pressure. Take moments about O. Moment of $dF = dF \times y = \rho g \sin\theta \, y^2 \, dA$.

$$F \times y^* = \rho g \sin\theta \int y^2 \, dA = \rho g \sin\theta \, I_O$$

where I_O = second moment of area about O. By the parallel-axis theorem, $I_O = I_G + A\,\bar{y}^2$. With $y = h/\sin\theta$ and $\bar{y} = \bar{h}/\sin\theta$:

$$\rho g\, A\, \bar{h} \times h^*/\sin\theta = \rho g \sin\theta \, (I_G + A\,\bar{h}^2/\sin^2\theta)$$

$$\boxed{\mathbf{h^* = \bar{h} + I_G \sin^2\theta \,/\, (A\,\bar{h})}}$$

For a **vertical** plane ($\theta = 90°$): $h = \bar{h} + I_G/(A\,\bar{h})$. *The centre of pressure always lies below* the centroid.*

Q8

2 marks

> A 3.6 m by 1.5 m wide rectangular gate MN is vertical and is hinged at a point 0.15 m below its centre of gravity. The total depth of water is 6 m. What horizontal force must be applied at the bottom of the gate to keep it closed?

The gate (3.6 m high, 1.5 m wide) is at the bottom of 6 m of water: top at 2.4 m depth, centroid at $\bar{h} = 2.4 + 1.8 =$ **4.2 m**, bottom at 6 m. Hinge at $4.2 + 0.15 =$ **4.35 m** depth.

$A = 3.6 \times 1.5 = 5.4\ m^2$; $I_G = 1.5 \times 3.6^3 / 12 = 5.832\ m^4$

Total pressure $F = \rho g\, A\, \bar{h} = 1000 \times 9.81 \times 5.4 \times 4.2 =$ **222.49 kN**

Centre of pressure $h = \bar{h} + I_G/(A\,\bar{h}) = 4.2 + 5.832/(5.4 \times 4.2) = 4.2 + 0.257 = 4.457\ m$*

The water force acts **0.107 m below the hinge** (4.457 − 4.35), so it tends to rotate the bottom of the gate outward (open). A horizontal force P at the bottom (6 m depth, 1.65 m below the hinge) must balance this moment about the hinge:

$P \times 1.65 = 222.49 \times 0.107$

$P =$ **14.45 kN** (applied at the bottom, towards the water)

Q9

2 marks

> Define buoyancy, buoyant force, metacentre and metacentric height.

- **Buoyancy**: the tendency of a fluid to **push upward** on a body immersed in it.
- **Buoyant force (upthrust)**: the **upward force** exerted by the fluid on an immersed or floating body, equal to the **weight of fluid displaced** (Archimedes' principle); it acts through the **centre of buoyancy** (centroid of the displaced volume).
- **Metacentre (M)**: the point about which a floating body starts to **oscillate** when given a small angular displacement; it is the point where the line of action of the buoyant force in the tilted position cuts the original vertical axis of the body.
- **Metacentric height (GM)**: the distance between the **centre of gravity G and the metacentre M**. GM = BM − BG, where BM = I/V. GM > 0 → stable; GM = 0 → neutral; GM < 0 → unstable.